

The low-dimensional second-order Hydrogen uncertainty conjecture

Literature-already-answered note for arXiv:2012.12667

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Status. **The conjecture is false in dimensions 2 and 3.** Chen–Tang prove this in arXiv:2508.15221, Theorem 1.1; their first-spherical-harmonic test gives a strict counterexample. The remaining dimension 4 was subsequently answered affirmatively by Huang–Ye, arXiv:2510.00453, Theorem 1.2. Thus the later literature now settles the entire range $2 \leq N \leq 4$.

The source conjecture

For $u \in W^{2,2}(\mathbb{R}^N)$ set

$$Q_N[u] := \frac{\int_{\mathbb{R}^N} |\Delta u|^2, dx \int_{\mathbb{R}^N} |\nabla u|^2, dx}{\left(\int_{\mathbb{R}^N} \frac{|\nabla u|^2}{|x|}, dx \right)^2}.$$

Theorem 2.3 of arXiv:2012.12667 proves $Q_N[u] \geq (N+1)^2/4$ for $N \geq 5$, sharply. Remark 2.4 on source PDF page 5 conjectures that the same inequality holds for $2 \leq N \leq 4$.

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Theorem 2.3. *Let $N \geq 5$ and $u \in W^{2,2}(\mathbb{R}^N)$. Then the following inequality holds*

$$\int_{\mathbb{R}^N} |\Delta u|^2 dx \int_{\mathbb{R}^N} |\nabla u|^2 dx \geq \frac{(N+1)^2}{4} \left(\int_{\mathbb{R}^N} \frac{|\nabla u|^2}{|x|} dx \right)^2 \quad (2.6)$$

where the constant $\frac{(N+1)^2}{4}$ is optimal and it is attained for functions of the form $u(x) = \alpha(1 + \beta|x|)e^{-\beta|x|}$, $\beta > 0$, $\alpha \in \mathbb{C}$.

Remark 2.4. *We conjecture that (2.6) still holds for $2 \leq N \leq 4$.*

Theorem 2.4. *Let $N \geq 2$ and $u \in W^{2,2}(\mathbb{R}^N)$. Then the following inequality holds*

$$\int_{\mathbb{R}^N} |\mathcal{R}_2 u|^2 dx \int_{\mathbb{R}^N} |\mathcal{R}_1 u|^2 dx \geq \frac{(N+1)^2}{4} \left(\int_{\mathbb{R}^N} \frac{|\mathcal{R}_1 u|^2}{|x|} dx \right)^2. \quad (2.7)$$

Also, (2.7) is optimal with the best constant $\frac{(N+1)^2}{4}$, which is achieved for functions of the form $u(x) = \alpha(1 + \beta|x|)e^{-\beta|x|}$, $\beta > 0$, $\alpha \in \mathbb{C}$.

Source evidence: arXiv:2012.12667, PDF page 5 (unaltered crop).

The exact later counterexample

Chen–Tang’s Theorem 1.1 (supporting PDF page 5) proves that for $N \in \{2, 3\}$ the sharp constant $C(N) = \inf_{u \neq 0} Q_N[u]$ satisfies

$$\frac{(N-1)^4(N+3)^2}{4(N^2-2N+5)^2} \leq C(N) \leq \frac{N(N+4)(N^2-1)^2}{4(N^2-N+4)^2} < \frac{(N+1)^2}{4}.$$

The final strict inequality is a direct negative answer to the source conjecture.

1.2 Main results and some comments

The first result of this paper reads as follows.

Theorem 1.1. *For $N \in \{2, 3\}$, the sharp constant in (1.3) satisfies*

$$C(N) < \frac{(N+1)^2}{4}, \quad (1.8)$$

more precisely,

$$\frac{(N-1)^4(N+3)^2}{4(N^2-2N+5)^2} \leq C(N) \leq \frac{N(N+4)(N^2-1)^2}{4(N^2-N+4)^2} < \frac{(N+1)^2}{4}.$$

Remark 1.2. Here we present some comments about Theorem 1.1.

- (1) The key point of proving Theorem 1.1 is to choose a suitable test function $u_* \in C_c^\infty(\mathbb{R}^N)$ such that

$$\frac{\int_{\mathbb{R}^N} |\Delta u_*|^2 dx \int_{\mathbb{R}^N} |\nabla u_*|^2 dx}{\left(\int_{\mathbb{R}^N} \frac{|\nabla u_*|^2}{|x|} dx \right)^2} < \frac{(N+1)^2}{4}, \quad \text{for } N \in \{2, 3\}.$$

When $N = 4$, it seems not easy to choose such suitable test function, and thus we can not obtain (1.8). We conjecture that (1.8) also holds for $N = 4$.

- (2) From Theorem 1.1, we know that when $N \in \{2, 3\}$, if the extremal function of (1.3) exists, it must be non-radial, this can be called symmetry breaking phenomenon.

Later-answer evidence: arXiv:2508.15221, Theorem 1.1 on PDF page 5 (unaltered crop).

Transparent verification of the test

Take a degree-one spherical harmonic ϕ and

$$u_*(r\sigma) = re^{-r}\phi(\sigma).$$

Up to a harmless angular normalization this is $u_*(x) = x_1 e^{-|x|}$. It belongs to $W^{2,2}(\mathbb{R}^N)$ for $N \geq 2$; standard cutoff and smoothing also place it in the closure of C_c^∞ under the relevant Sobolev norm. Writing $K = \int_{\mathbb{S}^{N-1}} |\phi|^2 d\sigma$, spherical-harmonic identities give

$$\begin{aligned} \int |\Delta u_*|^2 &= K \left(\frac{\Gamma(N+2)}{2^{N+2}} + (N+1) \frac{\Gamma(N)}{2^N} \right), \\ \int |\nabla u_*|^2 &= K \frac{\Gamma(N+2)}{2^{N+2}}, \\ \int \frac{|\nabla u_*|^2}{|x|} &= K \left(\frac{\Gamma(N+1)}{2^{N+1}} + \frac{\Gamma(N-1)}{2^{N-1}} \right). \end{aligned}$$

The factor K cancels, and $\Gamma(t+1) = t\Gamma(t)$ yields

$$Q_N[u_*] = \frac{N(N+4)(N^2-1)^2}{4(N^2-N+4)^2}.$$

Consequently

$$Q_2[u_*] = \frac{3}{4} < \frac{9}{4}, \quad Q_3[u_*] = \frac{84}{25} < 4.$$

Either value alone disproves the three-dimension-range conjecture; together they display the symmetry breaking in both low dimensions.

Proof intuition

The radial extremizer $(1 + \beta r)e^{-\beta r}$ realizes the conjectured constant, so a counterexample must be nonradial. Spherical-harmonic decomposition separates the three quadratic forms defining Q_N . In the first nonradial sector, $u = rv(r)\phi_1(\sigma)$, choosing $v = e^{-r}$ turns every radial integral into a Gamma integral. In $N = 2, 3$ the angular-energy contribution enlarges the weighted denominator enough to push the quotient strictly below the radial value. This is precisely the symmetry-breaking mechanism isolated by Chen–Tang.

Proof of Theorem 1.1. From the work of Cazacu, Flynn and Lam [8, Proof of Theorem 2.3], we know that the sharp constant of the second order HyUP (1.3) satisfies

$$C(N) \geq \inf_{k \in \mathbb{N} \cup \{0\}} \frac{1}{\left[1 + \frac{4k}{(N+2k-3)^2}\right]^2} \frac{(N+2k+1)^2}{4}, \quad \text{for all } N \geq 2.$$

This together with Lemma 2.1 implies that, for $2 \leq N \leq 4$,

$$C(N) \geq \frac{(N-1)^4(N+3)^2}{4(N^2-2N+5)^2}.$$

Then, in order to prove the symmetry breaking phenomenon, that is, $C(N) < \frac{(N+1)^2}{4}$ for $N \in \{2, 3\}$, it is enough to show that there exists a function $u_* \in C_c^\infty(\mathbb{R}^N)$ such that

$$\frac{\int_{\mathbb{R}^N} |\Delta u_*|^2 dx \int_{\mathbb{R}^N} |\nabla u_*|^2 dx}{\left(\int_{\mathbb{R}^N} \frac{|\nabla u_*|^2}{|x|^2} dx\right)^2} < \frac{(N+1)^2}{4}, \quad \text{for } N \in \{2, 3\}. \quad (2.4)$$

Here, we choose a test function $u_*(x) = |x|e^{-|x|}\phi_1(\sigma)$, where ϕ_1 is defined by (2.3) above. Then from [8, Lemma 3.1] with $v_1 := e^{-r}$, we have

$$\begin{aligned} \int_{\mathbb{R}^N} |\Delta u_*|^2 dx &= |\mathbb{S}^{N-1}| \left[\int_0^\infty |v_1'|^2 r^{N+1} dr + (N+1) \int_0^\infty |v_1'|^2 r^{N-1} dr \right] \\ &= |\mathbb{S}^{N-1}| \left[\int_0^\infty e^{-2r} r^{N+1} dr + (N+1) \int_0^\infty e^{-2r} r^{N-1} dr \right] \\ &= |\mathbb{S}^{N-1}| \left[\frac{\Gamma(N+2)}{2^{N+2}} + (N+1) \frac{\Gamma(N)}{2^N} \right], \end{aligned}$$

also

$$\begin{aligned} \int_{\mathbb{R}^N} |\nabla u_*|^2 dx &= |\mathbb{S}^{N-1}| \int_0^\infty |v_1'|^2 r^{N+1} dr \\ &= |\mathbb{S}^{N-1}| \int_0^\infty e^{-2r} r^{N+1} dr \\ &= |\mathbb{S}^{N-1}| \frac{\Gamma(N+2)}{2^{N+2}}, \end{aligned}$$

and

$$\begin{aligned} \int_{\mathbb{R}^N} \frac{|\nabla u_*|^2}{|x|} dx &= |\mathbb{S}^{N-1}| \left(\int_0^\infty |v_1'|^2 r^N dr + \int_0^\infty |v_1|^2 r^{N-2} dr \right) \\ &= |\mathbb{S}^{N-1}| \left(\int_0^\infty e^{-2r} r^N dr + \int_0^\infty e^{-2r} r^{N-2} dr \right) \\ &= |\mathbb{S}^{N-1}| \left[\frac{\Gamma(N+1)}{2^{N+1}} + \frac{\Gamma(N-1)}{2^{N-1}} \right]. \end{aligned}$$

Therefore,

$$\frac{\int_{\mathbb{R}^N} |\Delta u_*|^2 dx \int_{\mathbb{R}^N} |\nabla u_*|^2 dx}{\left(\int_{\mathbb{R}^N} \frac{|\nabla u_*|^2}{|x|} dx\right)^2} < \frac{\Gamma(N+2)}{2^{N+2}} + \frac{\Gamma(N)}{2^N} < \frac{\Gamma(N+1)}{2^{N+1}} + \frac{\Gamma(N-1)}{2^{N-1}}.$$

Chen–Tang’s explicit test and Gamma-integral computation, arXiv:2508.15221, PDF page 9 (unaltered crop).

The remaining dimension

The above test gives $Q_4[u_*] = 225/32 > 25/4$, so it does not decide $N = 4$. Huang–Ye subsequently proved the positive result

$$Q_N[u] \geq \frac{(N+1)^2}{4} \quad (N \geq 4),$$

with equality exactly at $\alpha(1+\beta r)e^{-\beta r}$. Their Theorem 1.2 therefore closes the last case left by Chen–Tang.

where the constant $\frac{(N+1)^2}{4}$ is optimal, and is attained by $u(x) = \alpha(1+\beta r)e^{-\beta r}$, with any $\beta > 0$, $\alpha \in \mathbb{R}$.

In other words, let $\|\cdot\|_2$ denote the standard norm in $L^2(\mathbb{R}^N)$ and

$$(1.7) \quad \mathcal{H}_N := \inf_{u \in W^{2,2}(\mathbb{R}^N), u \neq 0} \frac{\|\Delta u\|_2 \|\nabla u\|_2}{\int_{\mathbb{R}^N} \frac{|\nabla u|^2}{|x|^2} dx},$$

Cazacu–Flynn–Lam established that $\mathcal{H}_N = \frac{N+1}{2}$ for $N \geq 5$. Moreover, they conjectured that (1.6) holds for $2 \leq N \leq 4$. As the equality in (1.6) holds with $u(x) = \alpha(1+\beta r)e^{-\beta r}$, $\beta > 0$ in any dimension $N \geq 2$, then (1.6) will be sharp whenever it is valid.

Very recently, Chen–Tang give a negative answer to the above conjecture for dimensions two and three. More precisely, using explicit test functions, they obtain in [4]

$$\frac{1}{2} \leq \mathcal{H}_2 \leq \frac{\sqrt{3}}{2} \quad \text{and} \quad \frac{3}{2} \leq \mathcal{H}_3 \leq \frac{2\sqrt{21}}{5} \simeq 1.833.$$

So the remained question is for $N = 4$. Here we give a positive answer to (1.6) in $\mathbb{R}^4$, and present a unified study for $N \geq 4$.

Theorem 1.2. *For $N \geq 4$, then $\mathcal{H}_N = \frac{N+1}{2}$ and the extremal functions are*

$$u(x) = \alpha(1+\beta r)e^{-\beta r}, \quad \beta > 0 \text{ and } \alpha \in \mathbb{R}.$$

For the inequality in (1.6), we have $(\alpha_i) = (4-N, 2-N, 3-N)$. As explained above, to get the best constant $\mathcal{H}_N^2$ for (1.6), we can consider

$$(1.8) \quad \mathcal{H}_N^+ = 2\mathcal{H}_N = \inf_{u \in W^{2,2}(\mathbb{R}^N), u \neq 0} \frac{\|\Delta u\|_2^2 + \|\nabla u\|_2^2}{\left\| \frac{1}{\sqrt{r}} \nabla u \right\|_2^2}.$$

Another ingredient of our study is based on the link between Hardy inequalities and HUP. A notable example is: Using the famous Hardy inequality

Dimension-four closure: arXiv:2510.00453, Theorem 1.2 on PDF page 3 (unaltered crop).

Human review. Check Remark 2.4 on source PDF page 5, Chen–Tang Theorem 1.1 on supporting PDF page 5 and its test computation on pages 9–10, and Huang–Ye Theorem 1.2 on its PDF page 3.

References

- [1] C. Cazacu, J. Flynn, and N. Lam, *Sharp second order uncertainty principles*, arXiv:2012.12667.
- [2] X.-P. Chen and C.-L. Tang, *On the extremal functions of second order uncertainty principles: symmetry and symmetry breaking*, arXiv:2508.15221.
- [3] X. Huang and D. Ye, *On Sharp Heisenberg Uncertainty Principle and the stability*, arXiv:2510.00453.